# Projected Health Impact of Achieving WHO Targets for Cardiovascular Risk-Factor Control

Work in Progress

WHO · University of Washington

November 2026

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# Study Overview

# Aim 1: Research Questions

## Central Question

What is the projected **burden-of-disease impact** of achieving the WHO cardiovascular risk-factor control targets by **2030**?

**Sub-questions driving the analysis:**

1. How many **cardiovascular events** could be averted?
2. How many **CVD deaths** could be prevented by 2030 and beyond?
3. How would **life-years, YLLs, YLDs, and DALYs** change under joint achievement of:
   - 150M additional people with controlled hypertension
   - $\geq$50% lipid-lowering therapy coverage
   - 80% BP control among people with diabetes
   - 80% validated BP device availability

## Scope

**186 countries** · Ages 20–95 · 2026–2050 · Multiple CVD causes · WHO Target Scenarios

## Intervention start: 2026

Model calibrated to 2019 baseline. First year of target scale-up effects entering the projection is **2026**.

# Study Design at a Glance

| Component | Detail |
|---|---|
| **Model type** | Discrete-time multi-state model (Well → Sick → Dead) |
| **Calibration data** | WHO GHE 2023 CVD deaths (GBD 2023 where unavailable) & GBD 2023 prevalence; GBD 2023 input rates |
| **Population data** | UNWPP 2024 single-year age/sex projections |
| **Intervention start** | **2026** — first year intervention effects enter the projection |
| **Scale-up period** | Linear scale-up 2026 → 2030; rates held constant 2030–2050 |
| **Comparison** | Business-as-usual: 2024 control rates, no additional scale-up |
| **Outcome** | CVD deaths averted vs. BAU; age-standardised mortality rate (ASMR) |
| **Time horizon** | 2026–2050 (25-year horizon); model calibrated from 2019 |

# Study Context

**Objective**

- Quantify CVD deaths avertable by achieving WHO 2030 cardiovascular targets
- Coverage: **186 countries**, 2026–2050
- Four causes: IHD, ischemic stroke, hemorrhagic stroke, hypertensive HD

**Model**

- Discrete-time state transition (Well $\rightarrow$ Sick $\rightarrow$ Dead)
- Calibrated to WHO GHE 2023 deaths; projected with UNWPP 2024 populations
- 8 blood-pressure bins
- Baseline treatment effects explicitly de-duplicated

## Research Question

What are the projected deaths averted from achieving WHO hypertension control and lipid-lowering therapy targets across **186 countries**, 2026–2050?

## Pending △

Confirm primary metric: deaths averted vs. DALYs vs. LYG

# Data Sources

# Data Sources and Key Variables

| Source | Variables extracted | Role in model |
|---|---|---|
| **GBD 2023 (IHME)** | IR, CF, prevalence, background mortality (age/sex/country) | Baseline transition rates; initial states |
| **GBD 2019 (IHME)** | Relative risks per 10 mmHg SBP increase (IHD, stroke, HHD, by age) | Anchor BP-bin incidence; normalisation factor $\alpha$ |
| **Ettehad et al. 2016** | Trial effect sizes per BP category; diabetes-stratified | Treatment effect sizes by BP bin $\times$ cause |
| **UNWPP 2024 (UN)** | Population projections 2019–2050 (single-year age, sex, country) | Cohort sizes; rate denominators |
| **NCD-RisC** | Baseline HTN control rates ($C_0$); BP distribution by country | $C_0$; $P(\text{BP}_{\text{cat}})$ |
| **Country-specific** | HTN control targets 2030 (Progress / Ambitious / Aspirational) | $C_{\text{target}}$ per scenario |

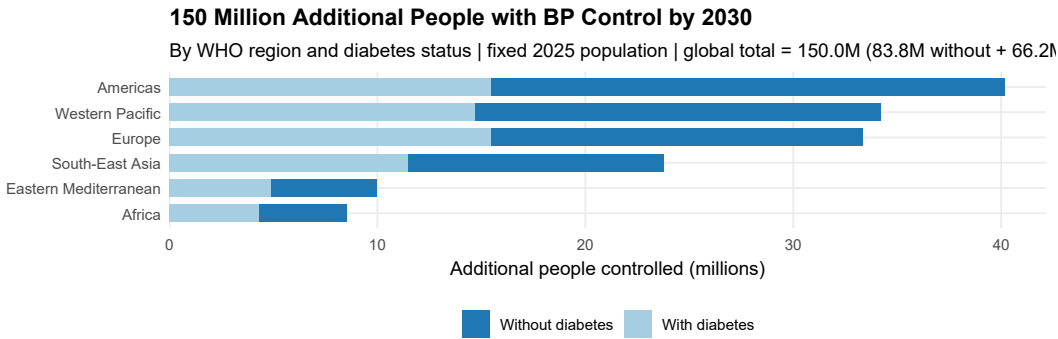
# Scenarios

# Four WHO Priority Targets

| # | WHO Target | Modelled Scenario |
|---|---|---|
| 1 | 150M more with HTN controlled by 2030 | **HTN Control (Combined)**: 150.0M more controlled = 83.8M without + 66.2M with diabetes (mutually exclusive, additive) |
| 2 | $\geq$50% eligible adults on lipid-lowering therapy by 2030 | **Improved Statin Uptake**: 50% coverage by 2030 |
| 3 | 80% BP control among diagnosed diabetics by 2030 | **HTN Control (Diabetes)**: 0.80 aspirational ceiling by 2030 (66.2M of the 150M) |
| 4 | 80% availability of validated BP devices | Operationally embedded in HTN scenarios (150M Target [1+3]) |

**Combination rule:** the two BP subgroups combine *additively* as a population mixture, $\varepsilon_{IR,\text{BP}} = 1 - (1 - w_D)e_N - w_D e_D$; the BP composite then combines multiplicatively with statins, $\varepsilon_{IR,\text{total}} = \varepsilon_{IR,\text{BP}} \times \varepsilon_{IR,\text{statin}}$.

# The 150 Million Target — Additional BP Control by Region and Diabetes Status



# HTN Control Target: 150 Million by WHO Region

| Region | HTN pop 2025 (M) | Controlled 2025 (M) | Cov. 2025 | Controlled 2030 (M) | Cov. 2030 | Added (M) | +% |
|---|---|---|---|---|---|---|---|
| Africa | 182.0 | 20.6 | 11.3% | 29.1 | 16.0% | 8.5 | 41.5% |
| Americas | 273.9 | 100.4 | 36.7% | 140.5 | 51.3% | 40.1 | 40.0% |
| Eastern Mediterranean | 157.5 | 24.0 | 15.2% | 34.0 | 21.6% | 10.0 | 41.5% |
| Europe | 300.3 | 80.5 | 26.8% | 113.9 | 37.9% | 33.3 | 41.4% |
| South-East Asia | 425.8 | 57.3 | 13.4% | 81.1 | 19.0% | 23.8 | 41.5% |
| Western Pacific | 456.1 | 83.5 | 18.3% | 117.7 | 25.8% | 34.2 | 41.0% |
| **Global** | **1,795.5** | **366.2** | **20.4%** | **516.2** | **28.8%** | **150.0** | **41.0%** |

Coverage rates and 2030 counts use the fixed 2025 hypertensive population; regional additions sum to exactly 150 million.

# HTN Control Target: 150 Million by WHO Region *and* Diabetes Subgroup

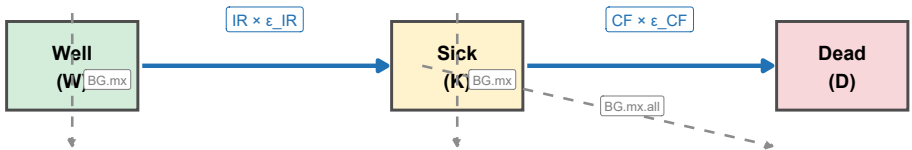
| | Additional controlled 2030 (M) | | | | |
|---|---|---|---|---|---|
| **Region** | **No diabetes** | **Diabetes** | **Total** | **Cov. 2030** | **+%** |
| Africa | 4.3 | 4.3 | 8.5 | 16.0% | 41.5% |
| Americas | 24.7 | 15.5 | 40.1 | 51.3% | 40.0% |
| Eastern Mediterranean | 5.1 | 4.9 | 10.0 | 21.6% | 41.5% |
| Europe | 17.9 | 15.4 | 33.3 | 37.9% | 41.4% |
| South-East Asia | 12.3 | 11.5 | 23.8 | 19.0% | 41.5% |
| Western Pacific | 19.5 | 14.7 | 34.2 | 25.8% | 41.0% |
| **Global** | **83.8** | **66.2** | **150.0** | **28.8%** | **41.0%** |

Additional people with BP control by 2030, disaggregated by WHO region *and* mutually exclusive diabetes status (fixed 2025 population). Both the regional totals and the two subgroup totals sum to exactly 150 million.

# Mathematical Model

# Model Architecture: Three Health States

Multi-State Transition Model | Annual time step | Ages 20-95 | 4 CVD causes




**Key:** $\varepsilon_{IR}, \varepsilon_{CF}$ = multiplicative effect ratios from the intervention module · BG.mx = background mortality · COVID-19 excess mortality applied 2020–2021 only

# Modeling Assumptions

## Model structure

- Discrete-time annual step (2019–2050); ages 20–95
- Four causes modeled independently
- COVID-19 excess mortality applied 2020–2021 only; no long-term effects

## Intervention mechanics

- Linear scale-up 2026 $\rightarrow$ 2030; held constant 2030–2050
- Only incremental coverage above $C_0$ (or $U_{0,\ell}$) generates new effects (no double-counting)
- Full adherence assumed for HTN control, statins adherence adjusted down for primary and secondary prevention.
- Statin effects restricted to ages $\geq 40$ and IHD & ischemic stroke only

## Blood pressure & effect sizes

- Static log-normal BP distribution: no secular trend, no endogenous updating as coverage rises
- Treatment eligibility: $\geq$140 mmHg (5 hypertensive bins)
- GBD RRs anchor bin-level baseline incidence
- Hypertensives split into mutually exclusive no-diabetes and diabetes subgroups; a person with diabetes is counted only in the diabetes subgroup
- Baseline control assumed 15 percentage points higher with diabetes, preserving the observed overall rate: $c_N = c - w_D(0.15)$, $c_D = c + (1 - w_D)(0.15)$
- Subgroup-specific Ettehad effect sizes applied to each subgroup's own share, then combined *additively* as a population mixture (no cross-term)

## Combination of interventions

- The two BP subgroups combine **additively**:

# Burden of Disease: Methods

**Three burden-of-disease metrics**

- **YLL** (Years of Life Lost): deaths $\times$ remaining life expectancy at age of death
- **YLD** (Years Lived with Disability): prevalent CVD cases $\times$ disability weight
- **DALY** (Disability-Adjusted Life Years): $DALY = YLL + YLD$

**DALYs averted** = $DALY_{\text{BAU}} - DALY_{\text{intervention}}$

Full derivation in Supplemental Mathematical Details.

## Reference standards

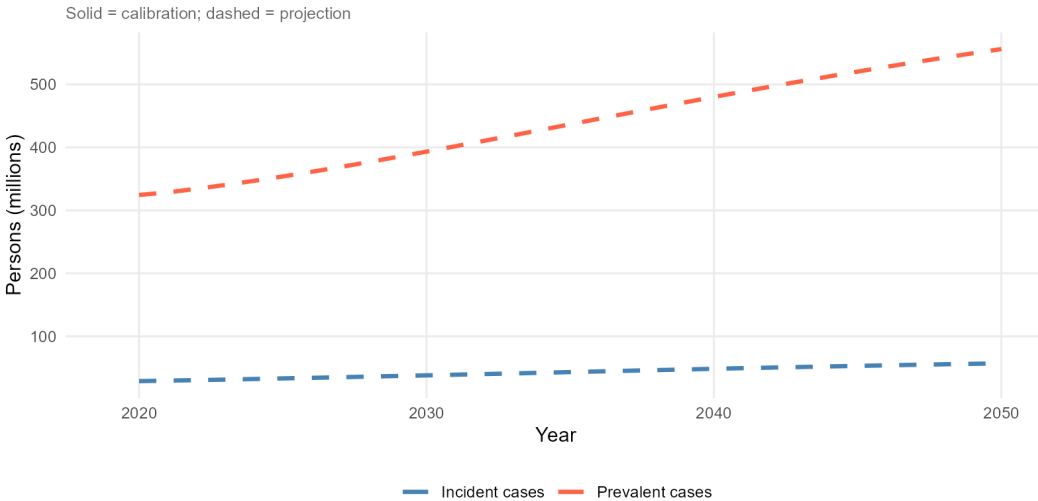
**Life table:** UNWPP 2024, medium variant (country- and year-specific $e_x$)

**Disability weights:** GBD 2023, derived as YLD/Prevalence per cause and location

**Discounting:** None (undiscounted)
**Age weighting:** None

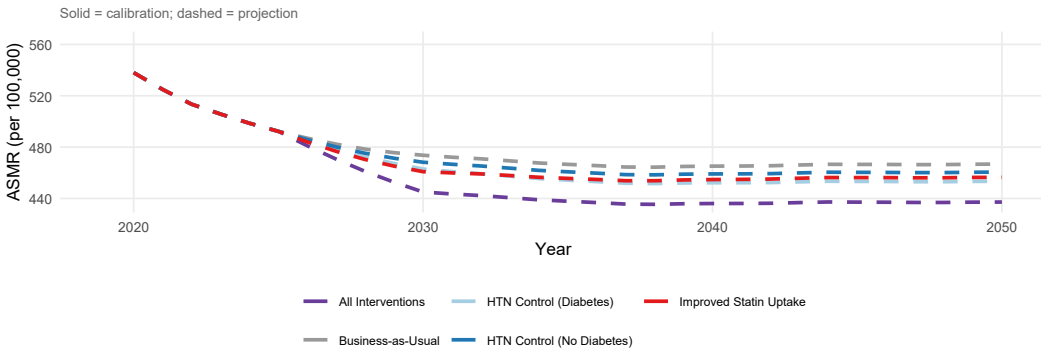
# Results

# Baseline Disease Trends (Business-as-Usual)




Incident and prevalent CVD cases 2019–2050; solid = calibration, dashed = projection.

# Age-Standardised CVD Mortality Rate



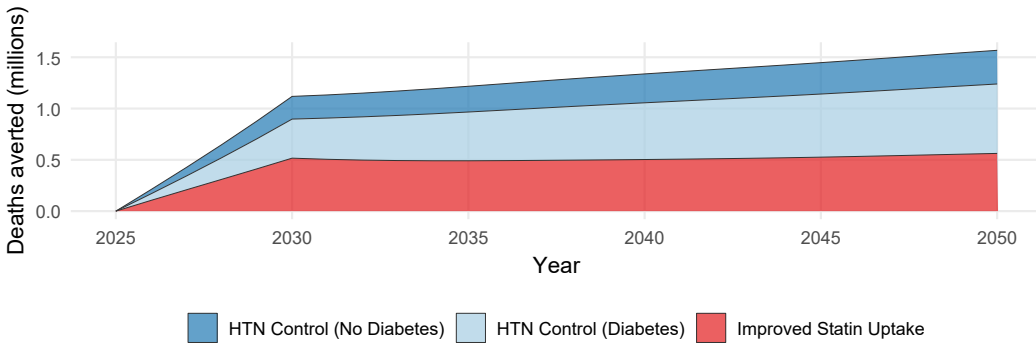
# Summary: Cumulative Deaths Averted, 2026–2050

| Scenario | Cumulative (M) | Avg./year (M) | % of BAU deaths |
|---|---|---|---|
| HTN Control (No Diabetes) | 6.3 M | 0.3 M | 1% |
| HTN Control (Diabetes) | 12.1 M | 0.5 M | 2% |
| Improved Statin Uptake | 11.8 M | 0.5 M | 1.9% |
| **All Interventions** | **30.1 M** | **1.2 M** | **4.9%** |

## Headline

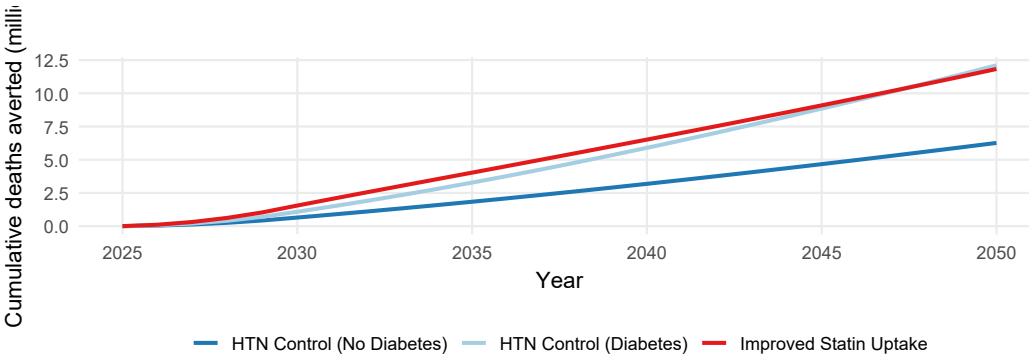
**All Interventions**: **30.1 million deaths averted** globally, 2026–2050 ($\approx$ 1.2 M/year).

# Annual Deaths Averted Over Time



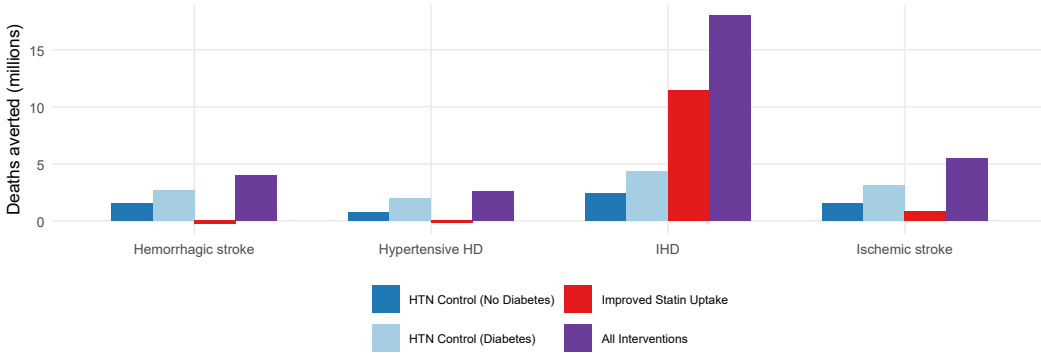

Cumulative deaths averted per intervention relative to business-as-usual.

# Cumulative Deaths Averted Over Time

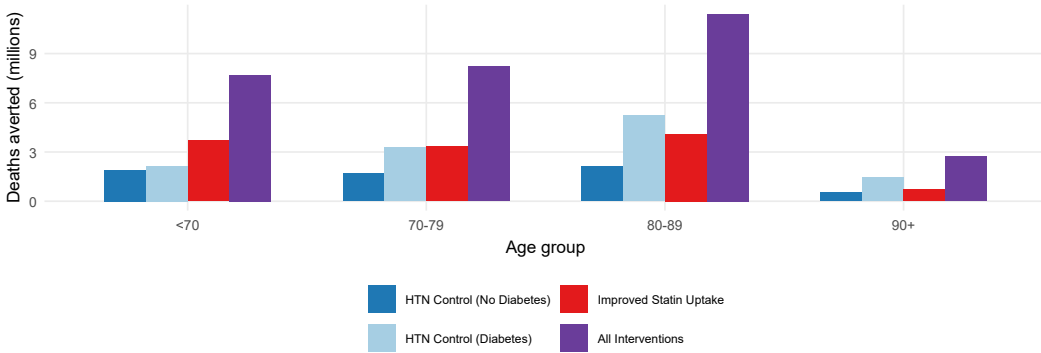



Cumulative deaths averted per intervention relative to business-as-usual.

# Deaths Averted by Cause of Death



# Deaths Averted by Age Group



# Burden of Disease Averted: YLLs, YLDs, and DALYs

- DALYs averted: 356.0 M
- YLLs averted: 351.4 M
- YLDs averted: 4.7 M

| Intervention | Deaths averted | DALYs averted | YLDs averted | YLLs averted |
|---|---|---|---|---|
| HTN Control (No Diabetes) | 6.3 M | 82.6 M | 1.9 M | 84.6 M |
| HTN Control (Diabetes) | 12.1 M | 121.4 M | 3.9 M | 125.4 M |
| Improved Statin Uptake | 11.8 M | 149.8 M | -1.2 M | 148.6 M |
| All Interventions | 30.1 M | 351.4 M | 4.7 M | 356.0 M |

## Headline

The combined WHO target package generates large health gains, with most benefits coming through avoided premature CVD deaths rather than morbidity reduction alone.

# Regional Summary

| Region | BAU deaths (M) | Averted (M) | % averted |
|---|---|---|---|
| Western Pacific | 207.6 | 9.9 | 4.8% |
| South-East Asia | 151.5 | 6.0 | 4% |
| Europe | 101.2 | 5.4 | 5.3% |
| Americas | 59.8 | 4.4 | 7.3% |
| Eastern Mediterranean | 53.8 | 2.5 | 4.7% |
| Africa | 42.5 | 1.8 | 4.3% |

## Observation

Absolute gains largest in high-burden regions; proportional gains reflect current coverage gaps.

# Economic Value of Avoidable Mortality

# Economic Valuation: Methods at a Glance

**Two complementary metrics**

- **Value of Statistical Life (VSL)** — monetary value the population places on small reductions in mortality risk; transferred from a U.S. reference using country GNI per capita (PPP) and an income elasticity.
- **Value of Statistical Life Year (VSLY)** — VSL annuitised by remaining life expectancy at the average adult age, then applied to age-specific life-years gained from deaths averted.

**Headline framing**: VSLY is reported as a **proportion of annual income** (discounted GNI) so results are directly comparable across regions and over time.

## Reference-case parameters

**Income elasticity (primary):** differential 0.8 (HIC) / 1.2 (LMIC), per Robinson et al. (2019)
**Sensitivity:** uniform $e = 1.5$
**Discount rate:** 3% (calendar-year)
**Income projection:** World Bank GNI (PPP), forward-projected with SSP2 GDP growth
**Horizon:** 2026–2050

## Methodological alignment

Approach follows **Robinson et al. (2019)** guidance and is consistent with recent published modelling of CVD policy benefits in World Bank's Healthy Longevity Initiative *Nature Medicine* papers (Chang et al. 2024; Verguet et al. 2024).

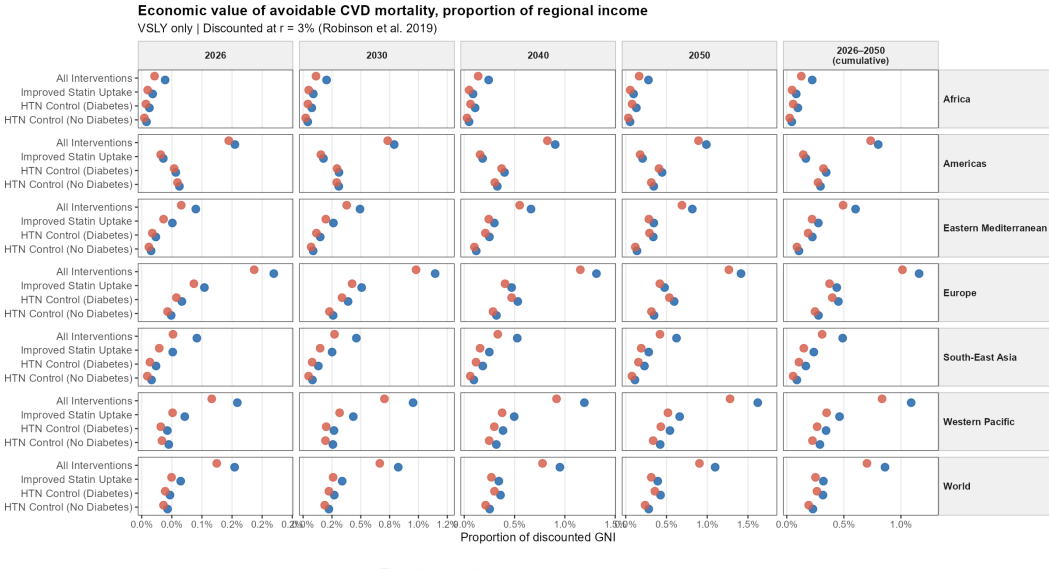
# Economic Value by WHO Region (Primary, Cumulative 2026–2050)

## World: All Interventions, 2026–2050 (primary, discounted at 3%)

**VSLY** = **0.9%** of annual income PPP (cumulative **$35.3 T**).

| Region | VSLY (USD T) | VSLY % GNI |
|---|---|---|
| Africa | $0.5 | 0.2% |
| Americas | $7.2 | 0.8% |
| Eastern Mediterranean | $1.8 | 0.6% |
| Europe | $10.5 | 1.2% |
| South-East Asia | $3.4 | 0.5% |
| Western Pacific | $12.0 | 1.1% |
| **World** | **$35.3** | **0.9%** |

# Value of Avoidable Mortality as a Proportion of Income



# Discussion

# Interpretation

**Key findings**

- **All Interventions** yields largest gains: HTN control + statins are complementary
- **IHD** and **ischemic stroke** drive the majority of avertable deaths
- Gains **accelerate after 2030** as coverage compounds
- **Older adults (70+)** benefit most in absolute terms
- **Top 15 countries** account for >75% of avertable burden

**Policy Implication**

Combined WHO Targets 1 + 2 are synergistic — neither alone captures the full benefit.

**Uncertainty △**

Sensitivity analysis to diabetes targeting assumptions and secular trends pending.

# Limitations

## Model structure & assumptions

- Stable transition rates assumed; no dynamic feedback between states
- Static BP distribution (log-normal): no secular trend and no endogenous updating as coverage increases
- No heterogeneity in adherence: newly HTN covered individuals assumed fully adherent
- COVID-19 excess mortality only modelled for 2020–2021; long-term effects excluded

## Data & parameters

- RCT-based effect sizes (e.g., Ettehad $\rho_{CF}$): real-world effectiveness at scale may be lower
- GBD 2023 uncertainty intervals not propagated
- Baseline $C_0$ from NCD-RisC; measurement error unquantified
- BP distributions rely on GBD modelling (not direct surveys in all countries)

## Scenarios & implementation realism

- Linear scale-up assumed (2026–2030); real programmes may follow non-linear trajectories
- No explicit health-system capacity constraints modelled
- Interactions with statins, diabetes care, and BP device availability not included

# Next Steps

**Analysis priorities**

1. Sensitivity analysis on effect sizes, trends and coverage
2. Add life-years gained (LYG) and DALYs as co-primary outcome
3. Stratify by income group (World Bank)
4. Conduct Aim 2: simulate alternative scale-up trajectories and implementation scenarios for hypertension control

Pending Decisions △

- Journal target?

# Summary

**1. The health gains are large and achievable**

All Interventions: **30.1 million deaths averted** globally, 2026–2050.

**2. HTN control + statins compound risk reduction**

HTN control and statins have multiplicative risk reductions, producing larger combined effects.

**3. Regional patterns reflect burden and coverage gaps**

Absolute gains largest in high-burden regions; proportional gains reflect current coverage gaps.

**4. The economic value rivals annual income itself**

World, All Interventions, 2026–2050: avoidable CVD mortality is worth **0.9% of annual income** (VSLY, cumulative **$35.3 T**, discounted at 3%).

*Meeting WHO 2030 CVD targets is not just aspirational — it is quantifiably lifesaving and economically transformative*

# Appendix

# Notation: Core Model

| Symbol | Description |
|---|---|
| *Core state-transition model* | |
| $a, s, c, t, \ell$ | Age, sex, cause, year, location |
| $W, K, D$ | Well, Sick, Dead populations |
| $IR_{a,s,c}$, $CF_{a,s,c}$ | Baseline incidence & case-fatality rates |
| $\varepsilon_{IR,j}$, $\varepsilon_{CF,j}$ | Multiplicative effect ratios for intervention $j$ |
| $BG.mx$, $BG.mx.all$ | Cause-specific & all-cause background mortality |
| $covid.mx$ | COVID-19 mortality (2020–2021 only) |

# Notation: HTN Control

| Symbol | Description |
|---|---|
| *HTN control (general & diabetes subgroup)* | |
| $C_0$, $C_t$, $C_{\text{target}}$ | Baseline, time-$t$, and target control rates |
| $\Delta C_{\text{target}}$, $\Delta C_{\text{increment}}$ | Total & time-varying incremental coverage |
| $C_{\text{agg}}^{\Delta}$ | Population-weighted incremental coverage (hypertensive bins) |
| $P(\text{BP}_{\text{cat}})$ | BP category population distribution (8 bins) |
| $RR_{\text{GBD}}$ | GBD relative risk per 10 mmHg SBP increase |
| $\alpha_{\text{inc}}$ | Incidence normalization factor |
| $E_D$, $E_N$ | Ettehad effect sizes (diabetes, no-diabetes subgroups) |
| $e_{g,t}$ | Baseline-adjusted incremental effect for subgroup $g \in \{N, D\}$ |
| $w_D$ | Diabetes share among hypertensives ($= P(D \mid H)$, mixture weight) |
| $c_N$, $c_D$ | Baseline control without / with diabetes ($c_D - c_N = 0.15$) |
| $\rho_{CF,c}$ | CF reduction factor per unit incremental coverage |
| $T_{\text{start}}$, $T_{\text{end}}$ | Scale-up start (2026) & end (2030) |

# Notation: Statins & Combined Effects

| Symbol | Description |
|---|---|
| *Statin therapy* | |
| $U_{0,\ell}$, $U_{\text{target}}$ | Baseline & target statin coverage |
| $\Delta U(t,\ell)$ | Incremental statin coverage above $U_{0,\ell}$ at time $t$ |
| $T_{\text{stat,start}}$, $T_{\text{stat,end}}$ | Statin scale-up start (2026) & end years |
| $RR_c^{IR}$, $RR_c^{CF}$ | Trial-based RRs for incidence & case fatality |
| $\delta_c^{IR}$, $\delta_c^{CF}$ | Risk reductions $= 1 - RR$ |
| $AF_{c,\ell}$ | Cause- & location-specific attributable fraction (incidence only) |
| $\psi_{\text{athero}}$ | Atherosclerotic fraction of ischemic stroke ($= 0.60$) |
| $E_{c,\ell}^{IR}(t)$, $E_{c,\ell}^{CF}(t)$ | Incremental effect sizes for $IR$ and $CF$ |

# HTN Intervention Module: 9-Step Pipeline

| | Action | Output |
|---|---|---|
| **1** | Baseline BP distribution $P(\text{BP}_{\text{cat}})$ without treatment (rx $= 0$) | $P(BP_{cat} \mid a, s, t)$ |
| **2** | GBD relative risks per 10 mmHg SBP $\rightarrow$ bin-specific incidence $IR_{\text{bin}}$ | $IR_{bin}$ |
| **3** | Subgroup-specific Ettehad effect sizes $E_N$, $E_D$ (no-diabetes, diabetes) per BP bin $\times$ cause | $E_{trial}$ |
| **4** | Linear coverage scale-up from $C_0$ to $C_{\text{target}}$, **starting 2026** | $C_t(t,\ BP_{cat})$ |
| **5** | Coverage adjustment: $E_{\text{adj}} = E_{\text{trial}}(C_t - C_0) \,/\, (1 - E_{\text{trial}}\, C_0)$ | $E_{adj}(t)$ |
| **6** | New bin incidence: $IR_{\text{bin,new}} = IR_{\text{bin}} \times (1 - E_{\text{adj}})$ | $IR_{bin,new}$ |
| **7** | Aggregate: $IR_{\text{new}} = \sum IR_{\text{bin,new}} \times P(\text{BP}_{\text{cat}})$ | $IR_{new}$ |
| **8** | Incidence effect ratio: $\varepsilon_{IR} = IR_{\text{new}} \,/\, IR_{\text{original}}$ | $\varepsilon_{IR} \rightarrow$ *model* |
| **9** | Case-fatality effect ratio: $\varepsilon_{CF} = 1 - \rho_{CF} \times C^{\Delta}_{\text{agg}}$ | $\varepsilon_{CF} \rightarrow$ *model* |

*Blue row = intervention start (2026). Green rows = outputs passed to state-transition model.*

# HTN Control: Coverage Scale-Up and Key Equations

**Incidence pathway**

*Step 2 — GBD RR anchors bin incidence:*

$$IR_{\text{bin}} = \frac{RR_{\text{GBD}} \times IR}{\alpha}, \quad \alpha = \sum_{\text{cat}} P_{\text{cat}} \cdot RR_{\text{GBD}}$$

*Step 5 — baseline-adjusted effect (no double-counting):*

$$E_{\text{adj}}(t) = \frac{E_{\text{trial}}\,(C_t - C_0)}{1 - E_{\text{trial}}\,C_0}$$

*Steps 7–8 — population-weighted effect ratio:*

$$\varepsilon_{IR} = \frac{\sum IR_{\text{bin,new}} \cdot P_{\text{cat}}}{IR_{\text{original}}}$$

**Case fatality pathway**

*Incremental coverage above baseline:*

$$\Delta C_\delta(t) = \max(C_t - C_0,\ 0)$$

*Aggregate across hypertensive bins (≥140 mmHg):*

$$C^{\Delta}_{\text{agg}} = \frac{\sum_{\geq 140} \Delta C_\delta \cdot P_{\text{cat}}}{\sum_{\geq 140} P_{\text{cat}}}$$

*CF effect ratio:*

$$\varepsilon_{CF} = 1 - \rho_{CF} \times C^{\Delta}_{\text{agg}}$$

| Cause | $\rho_{CF}$ |
|---|---|
| IHD | 0.24 |
| Ischemic stroke | 0.36 |
| Hemorrhagic stroke | 0.76 |
| HHD | 0.20 |

# HTN Control: Mutually Exclusive Diabetes / No-Diabetes Subgroups

**Two mutually exclusive, exhaustive subgroups** of the hypertensive population; each uses its own Ettehad effect size applied *only* to its own share.

*Subgroup-specific incremental effect ($g \in \{N, D\}$):*

$$e_{g,t} = \frac{E_g\,(c_{g,t} - c_{g,0})}{1 - E_g\,c_{g,0}}$$

*Baseline decomposition (preserves observed rate $c$):*

$$c_N = c - w_D(0.15), \qquad c_D = c + (1 - w_D)(0.15)$$

*Diabetes coverage scale-up toward the 0.80 ceiling:*

$$c_{D,t} = c_{D,0} + \Delta c_D \;\; \min\left[\tfrac{t-2026}{4},\, 1\right]$$

**Additive population mixture** ($w_D$ = diabetes share among hypertensives), *one* composite BP effect — no cross-term:

$$\varepsilon_{IR,\text{BP}} = 1 - (1 - w_D)\,e_{N,t} - w_D\,e_{D,t}$$

$$\Delta c_t = (1 - w_D)\,\Delta c_{N,t} + w_D\,\Delta c_{D,t}$$

## Why mutually exclusive?

People with diabetes have higher baseline control and larger DM-specific Ettehad effects. Counting each hypertensive once (diabetes *or* not) avoids double-counting and lets the two subgroup additions sum to exactly 150 million.

## Combined with statins

The BP composite then combines multiplicatively with the (independent) statin effect:

# Statin Therapy: Coverage Scale-Up and Key Equations

**Scope:** IHD & ischemic stroke; ages $\geq 40$ only

*Incremental coverage above baseline $U_{0,\ell}$:*

$$\Delta U(t,\ell) = \Delta U_{\text{target},\ell} \ \min\left[\frac{t - T_{\text{stat,start}} + 1}{T_{\text{stat,end}} - T_{\text{stat,start}} + 1}, 1\right]$$

where $\Delta U_{\text{target},\ell} = \max(U_{\text{target}} - U_{0,\ell},\ 0)$

*Trial-based RRs (per ≈1 mmol/L LDL-C reduction):*

| Cause | $RR^{IR}$ | $RR^{CF}$ |
|---|---|---|
| IHD | 0.74 | 0.80 |
| Ischemic stroke | 0.80 | 0.96 |

*Incidence (primary prevention, AF-scaled):*

$$E^{IR}_{c,\ell}(t) = \frac{AF_{c,\ell}\ \delta^{IR}_c\ \Delta U(t,\ell)}{1 - \delta^{IR}_c\ U_{0,\ell}}$$

*Case fatality (secondary prevention; no AF; atherosclerotic restriction for stroke):*

$$E^{CF}_{c,\ell}(t) = \begin{cases} \dfrac{\delta^{CF}_{\text{ihd}}\ \Delta U(t,\ell)}{1 - \delta^{CF}_{\text{ihd}}\ U_{0,\ell}}, & c = \text{ihd} \\ \dfrac{\psi_{\text{athero}}\ \delta^{CF}_{\text{ist}}\ \Delta U(t,\ell)}{1 - \delta^{CF}_{\text{ist}}\ U_{0,\ell}}, & c = \text{istroke} \end{cases}$$

$\delta^{IR}_c = 1 - RR^{IR}_c$; $\psi_{\text{athero}} = 0.60$
(atherosclerotic fraction)

*Effect ratios passed to model:*

$$\varepsilon_{IR,\text{stat}} = \begin{cases} 1 - E^{IR}_{c,\ell}(t), & a \geq 40,\ c \in \{\text{ihd,ist} \\ 1, & \text{otherwise} \end{cases}$$

$$\varepsilon_{CF,\text{stat}} = \begin{cases} 1 - E^{CF}_{c,\ell}(t), & a \geq 40,\ c \in \{\text{ihd,is} \\ 1, & \text{otherwise} \end{cases}$$

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# Thank You

Questions & Discussion

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